

GAJENDRASINH ZALA

Full Stack & AI Engineering | Java/J2EE, Spring Boot, GenAI-Assisted Development

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EXECUTIVE SUMMARY

Senior Full Stack Developer and AI Engineer with 12+ years designing and shipping production platforms for regulators, banks, payments networks, and global enterprises (MAS, OCBC, Visa, Toshiba, DHL). Strong backend engineering foundation in Java/J2EE (Spring Boot, Quarkus, Spring Security, Spring Batch, JPA/Hibernate), RESTful APIs, and microservices, paired with hands-on generative AI development — building an OpenAI-powered RAG chatbot with embeddings and a pgvector store in production, and using Claude Code CLI, GitHub Copilot, and Cline daily for code generation, code review, and automated test authoring, with additional hands-on experience across the Anthropic API and LangChain/LangGraph orchestration. Delivers scalable designs, maintainable code, and automated test coverage in regulated, high-availability, payments and financial-services environments, with a production-first mindset, strong ownership of code quality, and a track record of mentoring engineers on AI-enabled development practices.

TECHNICAL SKILLS

- **AI-Assisted Software Delivery:** Claude Code CLI, GitHub Copilot, Cline — daily use for code generation, code review, test generation, and documentation
- **Applied GenAI/LLM Engineering:** OpenAI API, RAG, embeddings, pgvector vector store, Anthropic API, LangChain/LangGraph orchestration, prompt engineering
- **AI & Machine Learning:** Python, Pandas, NumPy, ML pipeline design, AI-in-Finance analytics, model-assisted decisioning
- **Backend Engineering:** Java 8-20/J2EE, Spring Boot, Spring Security, Spring Batch, JPA/Hibernate, Quarkus, Node.js/Express/NestJS, Python/Django, RESTful APIs, microservices, Maven/Gradle
- **Full-Stack Engineering:** React.js (Hooks, Context API, Redux/Redux Toolkit), Angular, Next.js (SSR/SSG), LIT, TypeScript, JavaScript ES6, HTML5, CSS3, SASS/LESS, REST & GraphQL API design, micro-frontend architecture
- **Cloud & DevOps:** AWS (Lambda, ECS/EKS, EC2, S3, RDS, DynamoDB), Azure & Azure DevOps, Kubernetes/Amazon EKS, Docker, OpenShift, Jenkins CI/CD, Serverless Framework, CloudFormation, DevSecOps
- **Data & Messaging:** Kafka streaming, RabbitMQ, event-driven & reactive architecture, PostgreSQL (pgvector)/MySQL/Oracle/MongoDB/Redis/DynamoDB, TypeORM, schema design, query optimization, ETL and data migration
- **Testing & Quality:** TDD, Jest, Cypress, React Testing Library, unit & integration testing, code review practices
- **Monitoring & Observability:** ELK Stack, Prometheus, Grafana, AWS CloudWatch, production logging/alerting, incident response
- **Design Patterns & Architecture:** SOLID, DRY, clean architecture, event-driven & distributed systems, serverless, Agile/Scrum, technical mentoring
- **Domain Experience:** Financial services & banking, payments, POS & retail, regulatory/government-adjacent platforms, blockchain/CBDC, SAP integrations, compliance-driven delivery

KEY CAREER HIGHLIGHTS

- **GenAI RAG Platform (PathFin.ai):** Led full-stack delivery of an OpenAI RAG chatbot with embeddings and a pgvector store, plus a serverless AWS backend (Lambda/API Gateway/DynamoDB), cutting deployment time ~60% and infra cost ~40%, with 80%+ test coverage
- **CBDC Platform (MAS):** Architected a two-portal CBDC issuance platform on React/NestJS with upgradeable Solidity/OpenZeppelin ERC-20 contracts on a private Hyperledger Besu network, passing MAS security audits at ~99% compliance
- **Corporate Loan Approval Platform (OCBC):** Built 2 Java Spring Boot microservices and 2 React.js Micro Frontend (MFE) applications for a corporate loan approval system, with Kafka event-driven integration to downstream banking systems

- **High-Throughput Payments Engineering:** Designed microservices and GraphQL APIs on Spring Boot and AWS for Visa Worldwide, improving API throughput by 20% while maintaining stability at global payment-processing scale
- **Real-Time Retail & POS Systems:** Delivered real-time POS transaction microservices with Java, Quarkus, Spring Boot, and Kafka for Toshiba Global Commerce, supporting peak retail transaction load
- **SAP Integration Platform (Toshiba):** Led (Technical Lead) a 6-microservice Java/Spring Boot SAP-to-Fuel/POS integration platform processing ~1M records via batching and asynchronous processing, with a React.js operations portal

EMPLOYMENT HISTORY

- Senior Full Stack Developer & AI Engineer — Monetary Authority of Singapore (MAS) (via Manpower Staffing Services) | Nov 2025 – present
- Senior Full Stack Developer — Toshiba Global Commerce (via DNA Infotech) | Mar 2024 – Nov 2025
- Senior Full Stack Developer — OCBC Bank | Sep 2023 – Mar 2024
- Senior Software Engineer — Visa Worldwide (via NTT Singapore) | Oct 2022 – Aug 2023
- Software Engineer — DHL Information Services | DHL (Asia-Pacific) | Jun 2018 – Oct 2022
- Software Engineer / Java Developer — Plumslice Lab / Lumium Innovation | Internal Product Engineering | 2014 – 2018

PROFESSIONAL EXPERIENCE

Senior Full Stack Developer & AI Engineer, Monetary Authority of Singapore (MAS), Singapore | Nov 2025 – present

via Manpower Staffing Services (Singapore) Pte. Ltd.

Project: PathFin.ai — Financial AI Platform

- **Platform Architecture:** Accelerated AI-in-Finance adoption for PathFin.ai, measured by an 80%+ automated test coverage rate across all service layers, by architecting a full-stack platform on React/Vite, serverless AWS (Lambda, API Gateway, DynamoDB), and Python/Django.
- **OpenAI RAG Chatbot:** Delivered a production RAG-based conversational AI assistant, measured by secure, relevant responses to JWT/OTP-authenticated financial knowledge queries, by integrating OpenAI embeddings with a pgvector store for retrieval-augmented generation.
- **Serverless Cost & Speed:** Cut infrastructure cost ~40% and deployment time ~60%, measured against pre-migration baselines, by rebuilding the backend as auto-scaling serverless infrastructure provisioned via CloudFormation/Serverless Framework.
- **AI-Assisted Delivery:** Reduced feature delivery cycle time across two concurrent MAS projects, measured by daily shipped commits, by using Claude Code CLI and GitHub Copilot for AI-assisted code generation, code review, and automated test authoring.
- **Google Sheets Integration:** Cut content/event update times ~80%, measured by turnaround time for non-technical admins, by integrating the Google Sheets API for real-time data sync without code changes.
- **Mobile-Responsive Delivery:** Shipped a stable beta production release, measured by successful mobile-responsive rollout, by delivering a Vite/TypeScript/Tailwind CSS UI on Docker/Jenkins CI/CD.

Project: CBDC Platform — Full Stack Developer

- **Platform Architecture:** Delivered a two-portal CBDC request & issuance platform for MAS, measured by clean separation of external FI-facing and internal ops workflows, by architecting decoupled React (Next.js/Vite) frontends, NestJS microservice backends, and dedicated Web3 nodes on a private Hyperledger Besu network.
- **Smart Contracts:** Enabled secure, zero-downtime mint/burn for simulated FI transactions, measured by uninterrupted contract upgrades in production testing, by authoring upgradeable ERC-20 smart contracts in Solidity using OpenZeppelin's UUPS proxy pattern.
- **Security & Access Control:** Passed MAS security audits at ~99% compliance, measured by audit findings closure, by implementing OpenID-based SSO for external FI participants and Active Directory-based fine-grained authorization for internal ops users governing mint, burn, and audit workflows.
- **On-Chain Observability:** Cut transaction debugging time ~70%, measured by reduced investigation turnaround, by integrating a private Blockscout explorer for real-time on-chain monitoring and live audit trails.

- **Backend & Data:** Delivered a consistent, auditable data layer across dev/QA/prod, measured by schema integrity across environments, by designing RESTful APIs in Node.js/NestJS, modeling the relational schema in PostgreSQL with TypeORM, and containerizing services with Docker/Kubernetes/OpenShift.
- **Resilience & Compliance:** Delivered full traceability and regulatory-grade security across the CBDC lifecycle, measured by maker-checker control coverage, by establishing maker-checker controls, containerized CI/CD, centralized logging, and Grafana monitoring.

Senior Full Stack Developer, Toshiba Global Commerce, Singapore | Mar 2024 – Nov 2025

via DNA Infotech

Project: Real-Time POS Microservices

- **Real-Time POS Microservices:** Sustained real-time POS transaction throughput at peak retail load, measured by service resilience under load, by building resilient microservices with Java, Quarkus, Spring Boot, and Kafka.
- **Frontend Performance:** Cut frontend latency 25%, measured by dashboard load-time benchmarks, by optimizing React/GraphQL dashboards and adding Jest and Cypress regression coverage.
- **CI/CD & Container Orchestration:** Shortened deployment cycles, measured by release frequency, by building CI/CD pipelines on Kubernetes/OpenShift with Jenkins and Azure DevOps.
- **Caching & Query Performance:** Lowered query response times on high-traffic POS endpoints, measured by reduced average latency, by introducing Redis caching and query optimization.

Project: SAP Integration & Fuel Management Application — Full Stack Developer / Technical Lead

- **Platform Delivery:** Delivered a SAP-to-Fuel/POS integration platform from POC to production, measured by successful handoff across design, development, integration testing, performance tuning, and deployment, by owning the full lifecycle as Technical Lead.
- **Microservices Architecture:** Enabled independent scaling across SAP integration workflows, measured by decoupled service boundaries, by designing and building 6 Java/Spring Boot microservices for data synchronization, XML processing, transaction handling, and downstream communication.
- **Bi-Directional SAP Integration:** Delivered reliable bi-directional data flow between SAP and Fuel/POS systems, measured by structured integration workflow success rate, by implementing XML generation, validation, and processing for SAP-to-Fuel/POS and Fuel/POS-to-SAP synchronization.
- **High-Volume Data Processing:** Processed ~1 million records reliably, measured by successful batch-completion rate, by designing pagination, batching, and asynchronous processing for service scalability.
- **Operations Portal:** Improved visibility into integration workflows and transaction status, measured by reduced manual status-checking, by building a React.js operations portal integrated via REST APIs with the Spring Boot microservices.
- **Reliability Engineering:** Ensured reliable, consistent SAP/POS data exchange, measured by reduced integration failure rate, by applying exception handling, logging, validation, and retry mechanisms across integration services.

Senior Full Stack Developer, OCBC Bank, Singapore | Sep 2023 – Mar 2024

Project: Large Corporate Loan Approval Application

- **Platform Delivery:** Delivered an enterprise-grade Large Corporate Loan Approval platform end-to-end, measured by successful production rollout within a single 6-month engagement, by owning requirements analysis through deployment across 2 Spring Boot microservices and 2 React.js micro-frontends.
- **Micro Frontend Architecture:** Enabled modular, independently deployable frontend architecture for loan processing journeys, measured by decoupled release cycles across 2 applications, by designing and building 2 Micro Frontend (MFE) applications in React.js with reusable components, forms, validation, and state management.
- **Java Spring Boot Microservices:** Implemented core loan approval business logic reliably, measured by adherence to microservices and clean-code principles, by building 2 Java Spring Boot microservices using Spring MVC, REST APIs, JPA/Hibernate, and SQL.
- **Event-Driven Integration:** Improved decoupling and reliability between services and downstream banking systems, measured by reduced integration failures, by integrating Apache Kafka for asynchronous, event-driven communication.

- **Quality & Reliability:** Strengthened regression coverage across frontend and backend, measured by reduced defect-escape rate into production, by implementing robust error handling, validation, and logging, and building unit/integration test suites on React and Spring Boot components.
- **Build & Delivery:** Shortened build-to-deploy turnaround, measured by faster environment promotion, by using Docker and CI/CD pipelines, and led code reviews, defect resolution, and production issue analysis alongside BAs, architects, and QA.

Senior Software Engineer, Visa Worldwide, Singapore | Oct 2022 – Aug 2023

via NTT Singapore

- **Payment Platform Stability:** Sustained platform stability under peak payment-processing load, measured by consistent performance across regions, by developing on Angular, React, Spring Boot, and AWS with AI-assisted analytics.
- **API Throughput Optimization:** Improved API throughput 20% at global payment-processing scale, measured by sustained platform stability under peak load, by designing microservices and GraphQL APIs with standardized Maven-managed builds.
- **Release Automation:** Reduced release incidents, measured by fewer rollback events, by automating CI/CD workflows across Jenkins and Azure DevOps.
- **AI-Assisted Monitoring:** Surfaced anomalies in payment traffic patterns faster, measured by earlier detection in monitoring dashboards, by incorporating AI-assisted analytics into operational monitoring.
- **Cross-Functional Coordination:** Aligned API contracts and deployment schedules across regions, measured by reduced cross-team coordination friction, by collaborating with global payments teams.

Software Engineer, DHL Information Services, Asia-Pacific (Malaysia) | Jun 2018 – Oct 2022

- **Anomaly Detection Microservices:** Sped up anomaly detection in reconciliation data, measured by reduced manual review volume, by building Kafka/AWS/Redis-based microservices for automated data-matching.
- **Scalability & Architecture Upgrades:** Improved scalability for peak logistics volume, measured by sustained throughput during volume spikes, by leading architecture upgrades using JPA/Hibernate and Spring Batch.
- **Data Reconciliation Pipelines:** Reduced manual data-matching effort across logistics systems, measured by pipeline automation coverage, by building automated reconciliation and reporting pipelines.
- **Cloud Migration Support:** Improved processing throughput and reliability, measured by successful migration outcomes, by supporting migration of on-premises batch jobs to AWS-hosted services.
- **Cross-Regional Delivery:** Aligned feature delivery with regional logistics operating calendars, measured by consistent cross-region release cadence, by coordinating with Asia-Pacific stakeholders.

Software Engineer / Java Developer, Plumslice Lab Pvt Ltd & Lumium Innovation Pvt Ltd, India | 2014 – 2018

- **Core Java Development:** Maintained stable production operations, measured by defect turnaround time, by delivering core Java modules using Maven build tooling and SQL databases.
- **Application Support:** Ensured stable day-to-day production operations, measured by resolved-issue turnaround, by providing troubleshooting and defect-fix support for production applications.
- **Database Design:** Supported growing application data volumes without performance degradation, measured by query response times, by designing and optimizing SQL schemas and queries.
- **Foundational DevOps Exposure:** Built a foundation for later CI/CD work, measured by adoption of build automation and version control practices, through early exposure on production systems.

EDUCATION

- B.E. in Information Technology | Gujarat Technological University, India

CERTIFICATIONS

- Red Hat Application Development (Java EE)
- Red Hat DevOps Pipelines & CI/CD with Jenkins

INTERESTS

AI-Assisted Software Engineering, GenAI/LLM Application Development, Software Architecture, Cloud-native System Design, High-Performance Enterprise Applications, Mentoring Engineering Teams